

ReAlign: Text-to-Motion Generation via Step-Aware Reward-Guided Alignment

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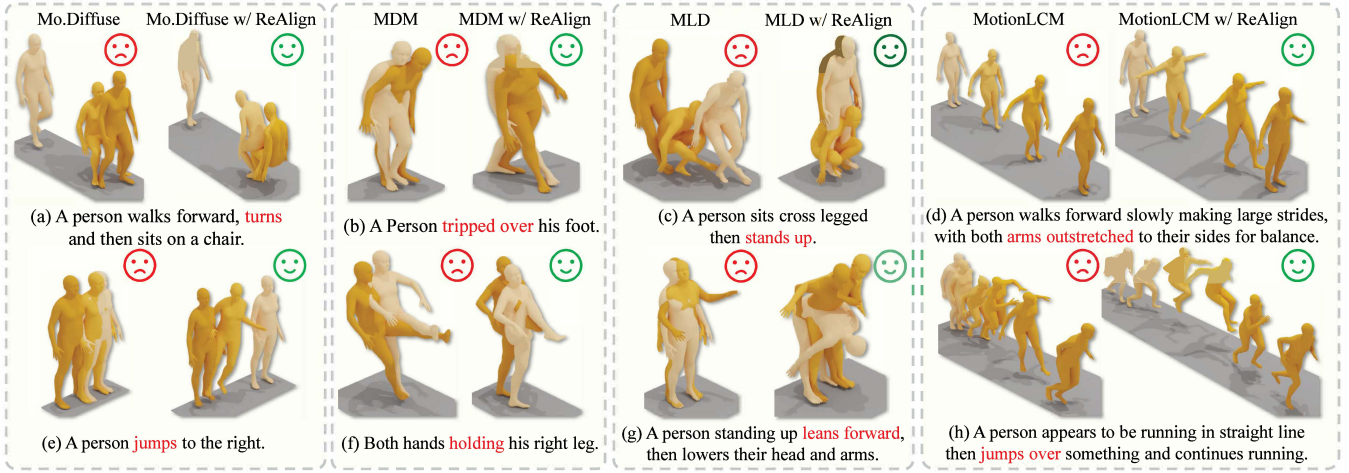


Figure 1: Visual comparison of text-to-motion generation. This figure presents motions generated by existing methods, such as Mo.Diffuse (2024), MDM (2023), MLD (2023b), and MotionLCM (2024). Our ReAlign enhances these models to produce motions that align more closely with text inputs.

Abstract

Text-to-motion generation, which synthesizes 3D human motions from text inputs, holds immense potential for applications in gaming, film, and robotics. Recently, diffusion-based methods have been shown to generate more diversity and realistic motion. However, there exists a misalignment between text and motion distributions in diffusion models, which leads to semantically inconsistent or low-quality motions. To address this limitation, we propose **Reward-guided sampling Alignment (ReAlign)**, comprising a step-aware reward model to assess alignment quality during the denoising sampling and a reward-guided strategy that directs the diffusion process toward an optimally aligned distribution. This reward model integrates step-aware tokens and combines a text-aligned module for semantic consistency and a motion-aligned module for realism, refining noisy motions at each timestep to balance probability density and alignment. Extensive experiments of both motion generation and retrieval tasks demonstrate that our approach significantly improves

text-motion alignment and motion quality compared to existing state-of-the-art methods.

Code — <https://wengwanjiang.github.io/ReAlign-page>

Introduction

With the rising demand for realistic 3D human motions in gaming, filmmaking, virtual reality, and robotics, along with recent advances in motion modeling (Jiang et al. 2023; Wang et al. 2025), there is an increasing need for intuitive and controllable motion generation techniques. Text-to-motion generation, which aims to synthesize human motion directly from natural language descriptions, has emerged as a key research topic (Chen et al. 2023b; Dai et al. 2024; Guo et al. 2022a; Jiang et al. 2023; Wu et al. 2025a,b).

Diffusion has emerged as the mainstream approach for text-driven motion generation (Tevet et al. 2023; Zhang et al. 2024). However, diffusion-based models often struggle with text-motion alignment due to their reliance on text embeddings encoded by CLIP (Radford et al. 2021), which is trained on text-image pairs rather than text-motion pairs. Consequently, these models often fail to capture the seman-

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tic alignment between text and motion. The motions synthesized by most existing diffusion-based methods lack coherence with the input descriptions (see Figure 1).

Prior works aiming to improve text-to-motion alignment, such as reinforcement learning with reward functions (Han et al. 2024; Tan et al. 2025), primarily focus on fine-tuning generative models to enhance motion quality. These approaches lack the capability to handle noisy motion inputs. Moreover, the misalignment issue should be addressed during the denoising process itself, rather than corrected retrospectively after the final motion is generated.

Besides semantic inconsistency, another practical limitation of existing methods is that the generated motions may still exhibit physically fragile behaviors, such as unstable contacts, foot skating, or motions that are difficult to track in downstream replay or simulation, even when they appear plausible at the perceptual level. This reveals a second gap in current text-driven motion generation: strong semantic alignment does not automatically guarantee strong physical executability.

Although existing diffusion-based methods can generate high-quality motions, they may still lack smoothness and realism in some cases. During reverse diffusion, motion generation relies solely on the generative model, without real motion references for guidance. This motivates us to turn to motion retrieval, which is closely related to motion generation but has rarely been unified with it in existing works. At the same time, semantic alignment and physical correction should not be treated in the same optimization space: the former mainly concerns whether the generated motion matches the textual intent, while the latter should focus on repairing physically vulnerable motion components without unnecessarily disturbing the semantics that have already been formed. Indeed, inappropriate physical correction may even alter the original motion structure and weaken the semantic expression conveyed by the generated motion.

To address these problems, we build upon the original reward-guided alignment design and further extend it toward physically robust motion generation. Specifically, we first retain the step-aware semantic alignment pathway through a Step-aware Reward Model (SPM), which derives a reward distribution from two complementary modules: a text-aligned module to ensure semantic consistency, and a motion-aligned module to assess motion realism. Together, these modules adapt to noisy motions and variations across timesteps, guiding the diffusion model toward a distribution that not only maximizes probability density but also maintains strong text-motion alignment. Building on this semantic guidance mechanism, we further introduce a *Step-aware Physical Proxy Model* (SPPM) to handle the physical side of the problem.

Unlike SPM, which operates as a semantic alignment signal over the denoising trajectory, SPPM performs selective repair in a constrained physical subspace of the motion representation. This design is important because physical correction should not rewrite the whole motion indiscriminately: it should only target the motion dimensions that are most closely related to physical stability and executability, while preserving the text-conditioned structure gener-

ated by the diffusion backbone. By coupling step-aware semantic guidance with semantic-preserved physical repair, the overall framework improves both text-motion coherence and physical robustness in a plug-and-play manner.

In this unified framework, SPM maintains the alignment of the generated motion with the input text, while SPPM selectively repairs physically fragile components during late denoising. The proposed design can be seamlessly integrated into existing motion diffusion models without re-training the backbone generator, providing a practical upgrade path from semantically plausible generation to semantically faithful and physically executable motion synthesis.

Our main contributions are as follows:

- **Step-aware semantic-preserved motion enhancement:** We propose a unified framework that improves diffusion-based text-to-motion generation by jointly considering semantic alignment and physical executability during denoising.
- **Complementary semantic and physical guidance:** We retain a step-aware semantic alignment pathway through SPM, and further introduce a Step-aware Physical Proxy Model that performs selective repair in a constrained physical subspace, improving physical robustness while preserving the original text-conditioned motion intent.
- **Plug-and-play enhancement of diffusion-based motion generation:** Extensive experiments demonstrate that our framework consistently enhances existing diffusion-based motion generation models in terms of both semantic fidelity and physical robustness.

Related Works

Alignment in Text-to-Motion Generation. Text-to-motion generation represents a critical task in computer vision, exhibiting rapid advancements in recent years (Guo et al. 2022a; Zhang et al. 2023a; Jiang et al. 2023; Zhang et al. 2025a; Yuan et al. 2024). Diffusion models have been adopted for text-driven motion generation (Tevet et al. 2023; Zhang et al. 2024). Representative diffusion-based approaches, including MDM (Tevet et al. 2023), MLD (Chen et al. 2023b), MotionDiffuse (Zhang et al. 2024), ReMoDiffuse (Zhang et al. 2023b), and MotionLCM (Dai et al. 2024), have improved motion diversity, latent-space generation, retrieval-augmented synthesis, and efficient controllable inference. Meanwhile, token-based and masked modeling methods, such as T2M-GPT (Zhang et al. 2023a), MoMask (Guo et al. 2024), and MoGenTS (Yuan et al. 2024), further enhance motion fidelity and discrete motion representation. Recent large-scale or unified motion models (Jiang et al. 2023; Zhang et al. 2025a; Wu et al. 2025b) also indicate a trend toward more general motion understanding and generation. Beyond generation backbones, cross-modal representation learning plays an important role in text-motion alignment. TEMOS (Petrovich, Black, and Varol 2022), TMR (Petrovich, Black, and Varol 2023), LaMP (Li et al. 2025c), and HumanTOMATO (Lu et al. 2024) improve text-motion representation, retrieval, captioning, or whole-body text alignment. These works provide stronger semantic priors for text-to-motion generation, but

they are mainly designed for kinematic text-motion consistency. A motion that is well aligned with text in the learned representation space may still exhibit physically fragile behaviors, such as unstable contact, foot skating, or poor trackability in downstream simulation.

Diffusion-Based Reward-Guided Generation. Alignment represents a versatile technique widely employed across the domains of language modeling (Rafailov et al. 2023), image generation (Wallace et al. 2024), and policy optimization (Chen et al. 2023a). Recently, reward- and preference-based alignment has also been studied in text-to-motion generation. ReinDiffuse (Han et al. 2025) optimizes motion diffusion with reinforcement learning for physically plausible rewards, while SoPo (Tan et al. 2025) combines online and offline preference optimization for improved generation quality. These methods demonstrate the importance of alignment beyond maximum-likelihood training, but they usually modify the generator through reinforcement learning, preference optimization, or other training-time objectives. Inference-time guidance in diffusion models can be driven by derivative-free sampling schemes or gradient-based surrogate scores. Derivative-free examples include Sequential Monte Carlo (SMC)-based guidance (Dou and Song 2024; Cardoso et al. 2024) and soft value-based decoding (Li et al. 2025a), while classifier guidance (Dhariwal and Nichol 2021) is a standard gradient-based mechanism. Inference-time guidance has also been studied for discrete diffusion models (Uehara et al. 2025; Nisonoff et al. 2025), while related search-based decoding algorithms in language models (Wan et al. 2024) show another route for improving inference-time alignment. Motion-specific guidance has also been explored for controllable synthesis: GMD (Karunratanakul et al. 2023) incorporates trajectory, keyframe, and obstacle constraints into motion diffusion, and Trace and Pace (Rempe et al. 2023) uses guided trajectory diffusion with a physics-based humanoid controller for pedestrian animation. Compared with these guidance methods, motion synthesis under both textual and physical objectives requires a more structured treatment because semantic alignment and physical correction do not necessarily share the same optimization space. Our reward-guided formulation preserves the plug-and-play, sampling-time nature of alignment, while the proposed physical proxy further extends this paradigm toward physically robust motion generation without re-training the diffusion backbone.

Physics-Aware Motion Generation and Control. Recent studies have increasingly recognized that visually plausible or semantically aligned motions are not necessarily physically executable. PhysDiff (Yuan et al. 2023) introduces physics-guided projection into diffusion-based human motion generation to alleviate artifacts such as floating, foot sliding, and ground penetration. ReinDiffuse (Han et al. 2025) refines motion diffusion through reinforcement learning to improve physical plausibility. CLoSD (Tevet et al. 2025) closes the loop between a diffusion planner and a reinforcement-learning-based tracking controller, enabling simulation-grounded character control. Morph (Li et al. 2025b) trains a physics-based refinement module with synthetic motions and uses the refined motions to improve

the generator, while POMP (Ji et al. 2025) further combines kinematic generation with simulation-based dynamics through phase manifolds to improve physical consistency. These methods demonstrate that physical executability has become an important objective beyond text consistency and visual realism. However, most existing physics-aware methods rely on simulator-coupled generation, policy fine-tuning, or specialized physical training pipelines. In contrast, our framework preserves the original diffusion backbone and extends step-aware semantic guidance with a Step-aware Physical Proxy Model, which selectively repairs physically fragile components in a constrained physical subspace during late denoising. This design provides a lightweight plug-and-play route for improving physical robustness while maintaining the semantic alignment established by the reward-guided sampling process.

Methods

Motivation and Overview Framework

Preliminaries. Existing diffusion-based motion generation methods (Chen et al. 2023b; Tevet et al. 2023) operate through a forward noising process and a reverse denoising process. The forward process gradually perturbs samples from the real motion distribution $p_{\text{data}}(\cdot)$ and can be written as a stochastic differential equation (SDE) (Song et al. 2021):

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}, \quad (1)$$

where t denotes the diffusion timestep, $\mathbf{f}(\cdot, \cdot)$ and $g(\cdot)$ are the drift and diffusion coefficients, and $\mathbf{w}$ is the Wiener process. The reverse process samples motions by following (Song et al. 2021):

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla \log p_t(\mathbf{x})]dt + g(t)d\mathbf{w}, \quad (2)$$

where $\nabla \log p_t(\mathbf{x})$ is the score of the learned sampling distribution.

For physical refinement, we additionally consider a kinematic motion $\mathbf{x} \in \mathbb{R}^{T \times D}$ and an offline simulator-tracker operator $\Pi(\cdot)$. This operator converts a HumanML3D motion into an Isaac-compatible humanoid reference, tracks it with a PHC-style controller (Luo et al. 2023), and maps the simulated rollout back to the normalized HumanML3D feature space. Each rollout provides a pair $(\mathbf{x}, \Pi(\mathbf{x}))$ and diagnostics $\mathbf{d}$, including tracking reward, reset state, root state, contact state, penetration, foot sliding, and joint errors. The tracking-reward diagnostics, such as reward and raw per-joint reward terms, follow the CLoSD simulator-controller reward convention (Tevet et al. 2025), while the remaining teacher fields are derived from our recorded rollout states and physical diagnostics. We use $\Pi(\cdot)$ only for offline supervision and replace it at generation time with a learned proxy. Let $M_{\text{phy}} \in \{0, 1\}^D$ denote the physically sensitive channels in the HumanML3D representation, including root height, relative joint coordinates, joint velocities, and foot-contact indicators.

Motivation. Text-to-motion diffusion models can synthesize plausible motions, but the learned sampling distribution $p_t(\mathbf{x}|c)$ is still dominated by motion likelihood. As shown in Figure 2, a sample prompted to walk forward to the right

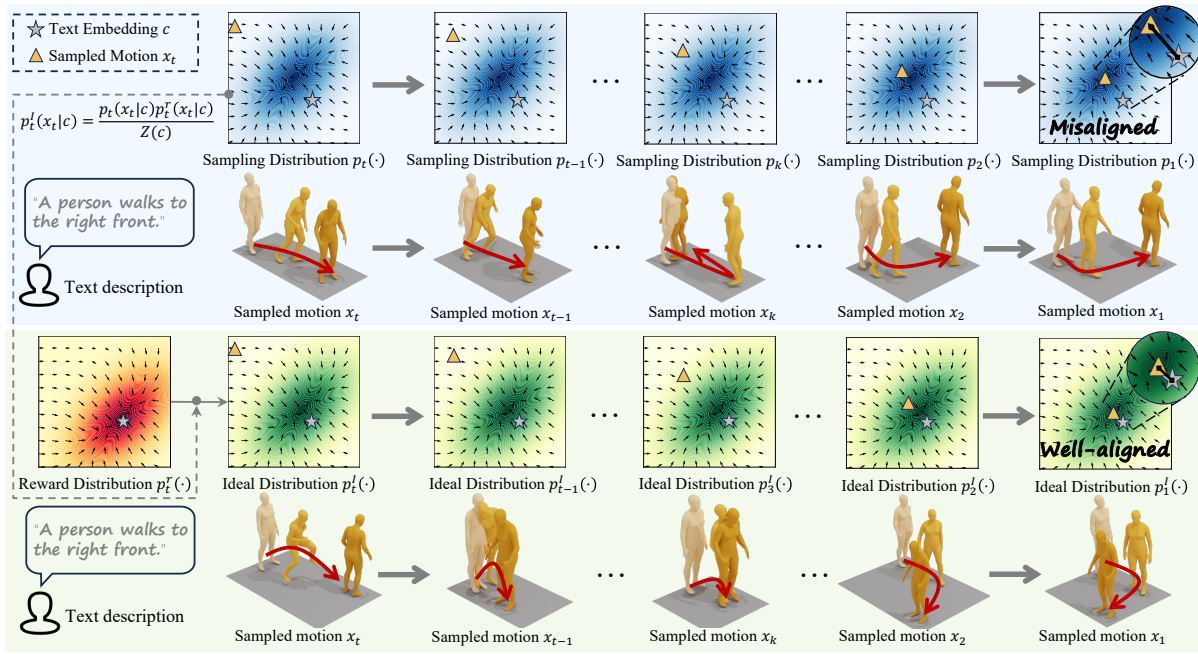


Figure 2: Illustration of the sampling process in diffusion-based motion generation frameworks. The blue region represents the sampling distribution $p_t(\cdot)$ learned by the diffusion model, while the green region depicts the ideal sampling distribution $p_t^I(\cdot)$ achieved by incorporating our proposed reward-guided sampling strategy with the sampling distribution $p_t(\cdot)$.

may be attracted to a high-density region that does not faithfully express the specified direction. This reveals a *semantic distribution gap*: a likely motion under the generator is not necessarily well aligned with the text condition.

A second gap appears when the generated motion is replayed by a physical character. Motions that look plausible in a kinematic representation may still contain unstable contacts, foot skating, ground penetration, or root trajectories that are difficult for a humanoid controller to track. This is a *physical executability gap*. Importantly, semantic alignment and physical executability should not be collapsed into a single unconstrained objective. Physical correction can alter gestures, limb usage, or left-right patterns that carry the text semantics. Therefore, our design separates the two roles: semantic reward guidance steers the denoising trajectory toward the intended meaning, while physical repair selectively modifies fragile physical components after the global motion structure has formed.

Overview Framework. We build a two-level inference-time alignment framework without retraining the diffusion backbone. The first level estimates a step-aware reward distribution $p_t^r(\mathbf{x}|c)$ from text-motion pairs and uses its gradient to guide reverse diffusion. We refer to this semantic branch as the step-aware semantic reward model (SPM). The second level introduces a Step-aware Physical Proxy Model (SPPM), which distills offline simulator rollouts into a bounded residual repair policy in the masked physical subspace. In this view, SPM changes the sampling trajectory in reward-gradient space, whereas SPPM performs a simulator-aware local projection in motion space and projects the result back to the latent trajec-

tory. The two branches are complementary: SPM preserves text-motion semantics, and SPPM improves physical executability through selective, confidence-weighted repair.

Formally, assume a reward distribution $p_t^r(\mathbf{x}|c)$ has been estimated. Then we define the ideal distribution as:

$$p_t^I(\mathbf{x}|c) = p_t(\mathbf{x}|c)p_t^r(\mathbf{x}|c)/Z(c), \quad (3)$$

where $Z(c) = \int p_t(\mathbf{x}|c)p_t^r(\mathbf{x}|c)d\mathbf{x}$ is a normalizing constant. This formulation integrates the original sampling distribution $p_t(\mathbf{x}|c)$ with the reward distribution $p_t^r(\mathbf{x}|c)$, balancing both probability density and text-motion alignment. The resulting semantic guidance is complementary to the physical proxy introduced later: it changes the sampling trajectory through the reward gradient, whereas SPPM performs a bounded residual update in motion space to improve physical executability while preserving the text-conditioned semantics.

Using this ideal distribution, we modify the reverse process for trading-off semantic alignment and high-probability sampling as stated in the following theorem.

Theorem 1. When using the ideal sampling distribution $p_t^I(\mathbf{x}|c)$ in Eq. (3) to replace the vanilla sampling distribution $p_t(\mathbf{x}|c)$, the reverse SDE becomes:

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla (\log p_t(\mathbf{x}|c) + \log p_t^r(\mathbf{x}|c)) \right] dt + g(t) d\mathbf{w}. \quad (4)$$

Theorem 1 shows that the gradient of the ideal sampling distribution decomposes into the gradients of $p_t(\mathbf{x}|c)$ and $p_t^r(\mathbf{x}|c)$. Since $p_t(\mathbf{x}|c)$ is already known, the estimated reward distribution can directly guide the sampling process

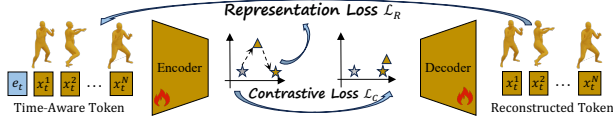


Figure 3: Framework of step-aware reward model. During this process, time-aware tokens, consisting of timestep embedding t and motion embeddings x_t^k , are aligned with text embedding c in the latent space and reconstructed via the decoder, with the encoder and decoder jointly optimized by contrastive loss $\mathcal{L}_C$ and representation loss $\mathcal{L}_R$ (Petrovich, Black, and Varol 2022).

toward the ideal distribution. Next, we detail the estimation of the reward distribution and outline the motion sampling procedure.

Step-Aware Alignment for Reward Distribution

A core challenge in estimating the reward distribution $p_t^r(\mathbf{x}|c)$ is achieving precise motion-text alignment under varying noise levels in the reverse diffusion process (Liang et al. 2025; Rempe et al. 2023). Existing methods (Petrovich, Black, and Varol 2023; Karunratanakul et al. 2023; Li et al. 2025c) assume clean and noise-free motion, and overlook timestep-dependent distortions, resulting in coarse and inconsistent alignments. This misalignment hinders accurate reward estimation, which is critical for guiding sampling toward semantically faithful motion generation. To address this, we introduce a step-aware reward model for noise-adaptive alignment and a motion-to-motion reward to ensure consistency with real-world motion patterns implied by text. These components are integrated into a unified reward distribution to enhance alignment and motion quality.

Step-Aware Reward Model. To mitigate timestep-dependent misalignment, we introduce a step-aware reward model $R(\cdot)_\varphi$ illustrated in Figure 3, which explicitly accounts for noise variations across diffusion timesteps. Unlike conventional alignment models (Petrovich, Black, and Varol 2023; Li et al. 2025c), our approach incorporates a timestep token $[e_t]$ into the motion representation, allowing the model to learn noise-dependent alignment patterns. Given an N -frame motion sequence $[x_t^1, x_t^2, \dots, x_t^N]$, we augment it with the timestep token to form the enriched representation $[e_t, x_t^1, x_t^2, \dots, x_t^N]$. This enables the transformer-based encoder to process motion dynamics while adapting to different noise levels.

During training, noise is added to motion at timestep t , and the step-aware reward model $R_\varphi(\mathbf{x}_t, c)$ is optimized by two complementary losses: a representation loss $\mathcal{L}_R$ (Petrovich, Black, and Varol 2022) to learn meaningful motion embeddings, and a contrastive loss $\mathcal{L}_C$ (Oord, Li, and Vinyals 2018) to ensure accurate motion-text retrieval. The overall training loss $\mathcal{L}_{RM}(\varphi; \mathbf{x}_t, c)$ is defined as:

$$\mathcal{L}_{RM}(\varphi; \mathbf{x}_t, c) = \mathcal{L}_C(\varphi; \mathbf{x}_t, c) + \mathcal{L}_R(\varphi; \mathbf{x}_t, c). \quad (5)$$

Algorithm 1 details the training procedure of the step-aware reward model.

Once trained, the step-aware reward model establishes a

Algorithm 1: Training Step-Aware Reward Model

Input: Step-aware semantic reward model R_φ , text-motion training set $\mathcal{D}_{tr}$, timestep range $[t_{\min}, t_{\max}]$, noising probability ω , noise schedule α .

Output: Step-aware reward model R_φ .

```

1: repeat
2:   for  $(\mathbf{x}, c)$  in  $\mathcal{D}_{tr}$  do
3:      $t \leftarrow 0$ 
4:     if  $\text{Uniform}(0, 1) < \omega$  then
5:        $t \leftarrow \text{Uniform}(t_{\min}, t_{\max})$ 
6:     end if
7:      $\mathbf{x}_t \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}, (1 - \alpha_t)\mathbf{I})$ 
8:     Encode  $[e_t, \mathbf{x}_t]$  and  $c$  into the shared semantic space.
9:     Compute  $\mathcal{L}_{RM}(\varphi; \mathbf{x}_t, c)$  by Eq. (5).
10:     $\varphi \leftarrow \varphi - \nabla_\varphi \mathcal{L}_{RM}(\varphi)$ .
11:  end for
12: until converged

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well-aligned latent space. Given a motion $\mathbf{x}$ and text condition c , it evaluates their semantic alignment as:

$$R_\varphi(\mathbf{x}, c) = \cos(\mathbf{z}_\mathbf{x}, \mathbf{z}_c), \quad (6)$$

where $\mathbf{z}_\mathbf{x}$ and $\mathbf{z}_c$ are the respective motion and text embeddings in the learned latent space.

Motion-to-Motion Reward. While text-to-motion alignment is essential, text descriptions often exhibit ambiguity, leading to inconsistencies in generated motions. To mitigate this, we introduce a motion-to-motion reward, which evaluates alignment by comparing the generated motion $\mathbf{x}_t$ with a reference motion $\mathbf{x}^c$ retrieved from the training set $\mathcal{D}_{tr}$. The step-aware reward model is used to select $\mathbf{x}^c$ as the closest match to the text condition c :

$$\mathbf{x}^c = \arg \max_{\mathbf{x} \in \mathcal{D}_{tr}} R_\varphi(\mathbf{x}, c). \quad (7)$$

This retrieved motion $\mathbf{x}^c$ acts as a dynamic anchor, ensuring that generated motions remain faithful to real-world motion patterns implied by the text. Accordingly, The motion-aligned reward is then computed as:

$$R_m(\mathbf{x}_t, c) = \cos(\mathbf{z}_\mathbf{x}, \mathbf{z}_{\mathbf{x}^c}), \quad (8)$$

where $\mathbf{z}_\mathbf{x}$ and $\mathbf{z}_{\mathbf{x}^c}$ are the embeddings of the generated and retrieved motions, respectively. This ensures generated motions adhere to real-world motion patterns while maintaining semantic consistency.

Reward Distribution. With both the step-aware reward model and the motion-to-motion reward, we define the dual-alignment reward as:

$$R(\mathbf{x}_t, c) = \mu R_\varphi(\mathbf{x}_t, c) + \eta R_m(\mathbf{x}_t, c), \quad (9)$$

where μ and η control the contributions of text-based and motion-based alignment. This reward formulation defines the reward distribution over noised motion as:

$$p_t^r(\mathbf{x}_t|c) = \exp(R(\mathbf{x}_t, c)) / Z^r(c). \quad (10)$$

Here, $Z^r(c) = \int \exp(R_\varphi(\mathbf{x}, c)) d\mathbf{x}$ is for normalization.

By integrating text-motion and motion-motion alignment, our approach constructs a robust reward signal that captures

both semantic consistency and motion coherence. This enables more precise guidance of the diffusion sampling process, ensuring that generated motions are not only probable but also faithful to their textual descriptions.

Reward-Guided Sampling

Building on the dual-alignment reward $R(\mathbf{x}_t, c)$ and its associated distribution $p_t^r(\mathbf{x}_t|c)$, we now integrate them into the reverse SDE to refine motion generation. The following theorem establishes how this reward distribution enhances sampling for precise text-conditioned synthesis.

Theorem 2. *Given the reward distribution $p_t^r(\mathbf{x}|c)$ defined in Eq. (10), the reverse SDE can be rewritten as:*

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla (\log p_t(\mathbf{x}|c) + R(\mathbf{x}_t, c)) \right] dt + g(t) d\mathbf{w}. \quad (11)$$

Theorem 2 reveals that the reward gradient $\nabla R(\mathbf{x}_t, c)$, derived from both text-aligned and motion-aligned reward components, directly influences the sampling trajectory. Integrating these gradients into the reverse SDE can dynamically steer the sampling toward a distribution that better aligns with both textual conditions and realistic structures.

Building upon this continuous-time formulation, for practical motion generation we then derive its discrete approximation within the DDPM (Ho, Jain, and Abbeel 2020) framework in the following theorem.

Theorem 3. *Given a reverse SDE defined in Eq. (11), adopting standard DDPM settings (Song et al. 2021; Ho, Jain, and Abbeel 2020) where $\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\bar{\beta}_{t+\Delta t}\mathbf{x}_t$, $g(t) = \sqrt{\beta_{t+\Delta t}}$, and $\bar{\beta}_t = \frac{\beta_{t+\Delta t}}{\Delta t}$, with time steps $N \rightarrow \infty$ and step size $\Delta t = \frac{1}{N}$, the reward-guided denoising process is given by:*

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\bar{\mathbf{x}}_{t-1} + \sqrt{\beta_t} \epsilon \right) + \frac{\beta_t}{\sqrt{\alpha_t}} \nabla R(\mathbf{x}_t, c), \quad (12)$$

where $\bar{\mathbf{x}}_{t-1} = \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\mathbf{x}_t, t, c)$, β_t and α_t are the noise schedule parameters, $\epsilon_\theta(\cdot)$ represents the diffusion model network, and ϵ is Gaussian noise sampled from $\mathcal{N}(\mathbf{0}, \mathbf{I})$.

Theorem 3 demonstrates that the reward gradient $\nabla R(\mathbf{x}_t, c)$, weighted by $\frac{\beta_t}{\sqrt{\alpha_t}}$, progressively influences the denoising process, adapting the sampling trajectory toward a distribution that reflects the intended motion semantics.

To ensure the sampling stability, we remove the weight $\frac{\beta_t}{\sqrt{\alpha_t}}$ on the reward term, leading to a revised denoising process:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\bar{\mathbf{x}}_{t-1} + \sqrt{\beta_t} \epsilon \right) + \nabla R(\mathbf{x}_t, c). \quad (13)$$

The semantic update above forms the reward-guided component of our generation process. Next, we introduce the physical proxy that repairs physically vulnerable components after the semantic structure has been established.

Step-Aware Physical Proxy Model

Physical objective. The reward-guided process in Eq. (13) improves semantic alignment by adding the gradient of a

learned reward distribution. However, this reward does not directly measure whether a motion can be executed by a simulated humanoid. Given the simulator-tracker operator $\Pi(\cdot)$ introduced above, a natural physical objective is to move a generated motion toward its simulator-consistent counterpart in the physically sensitive subspace:

$$\mathcal{E}_{\text{phy}}(\mathbf{x}) = \frac{1}{2} \|M_{\text{phy}}(\Pi(\mathbf{x}) - \mathbf{x})\|_2^2, \quad (14)$$

Directly optimizing Eq. (14) at every denoising step is impractical, since $\Pi(\cdot)$ requires SMPL/Isaac conversion and physics rollout. More importantly, $\Pi(\cdot)$ is not a semantic objective: blindly moving a sample toward the simulated trajectory may alter the text-conditioned gesture or left-right body semantics. Therefore, we distill the simulator into a step-aware proxy and use it as a *selective* repair controller rather than an unconstrained motion editor.

Offline physical teacher construction. We first build a physical teacher set by running an offline simulation pipeline. HumanML3D motions are filtered to remove sequences that are unsupported by the current full-body tracking setup, converted from 22-joint trajectories to SMPL parameters, and then transformed into Isaac-compatible humanoid motions. A PHC-style tracking controller (Luo et al. 2023) replays each reference motion and records the simulated trajectory together with diagnostic signals. The rollout is then mapped back to normalized HumanML3D features, giving a paired teacher sample

$$\mathbf{x}_{\text{in}} \xrightarrow{\Pi} \mathbf{x}_{\text{tar}}, \quad \Delta^* = \mathbf{x}_{\text{tar}} - \mathbf{x}_{\text{in}}. \quad (15)$$

This conversion gives SPPM supervision in the same representation used by the frozen motion generator, avoiding a mismatch between simulator coordinates and diffusion features. To reduce the gap between clean dataset motions and generated motions, we organize the teacher into three complementary views: clean reference replay for stable supervision, hard replay for failure-prone training motions, and generated-motion replay from the frozen base generator for deployment-style errors. The generated and hard-replay views are constructed from training prompts only, and checkpoint selection uses the clean validation split together with held-out monitors from the other two views. Thus, the proxy is not trained on test motions and is not selected only for clean reference imitation.

Canonical teacher factorization. Each simulator rollout contains heterogeneous signals, but SPPM should not treat them as a bag of unrelated supervision heads. We instead factorize the offline teacher into a decision-first physical repair target:

$$\mathcal{T}_\omega(\mathbf{d}, \Delta^*) = (\mathcal{D}^*, \mathcal{C}^*, \mathcal{B}^*, \Delta^*), \quad (16)$$

where $\mathcal{D}^*$ diagnoses *what is physically wrong*, $\mathcal{C}^*$ estimates *whether the simulator residual is trustworthy*, and $\mathcal{B}^*$ decides *where repair capacity may be spent*. This factorization is the central abstraction of SPPM: the proxy does not merely regress the simulator residual Δ^* ; it first learns a structured repair policy that determines when, where, and how strongly the residual can be applied.

Concretely, let $\mathbf{d}$ include tracking reward, body/root errors, contact state, penetration, foot sliding, reset state, and residual magnitude. The diagnostic factor summarizes global quality and local risk,

$$\mathcal{D}^* = [\mathcal{S}_\omega(\mathbf{d}), \mathcal{R}_\omega^f(\mathbf{d}, \Delta^*), \mathcal{R}_\omega^j(\mathbf{d}, \Delta^*)], \quad (17)$$

corresponding to sequence quality, frame risk, and joint risk. The reliability factor combines contact reliability, root feasibility, residual confidence, hard-failure state, and repairability into a single trust estimate:

$$\mathcal{C}^* = \mathcal{Q}_\omega(\mathbf{d}, \mathcal{D}^*, M_{\text{phy}}\Delta^*). \quad (18)$$

This term is necessary because a large simulated residual is not always a useful correction direction: after early reset or severe penetration, the rollout remains valuable for diagnosis, but its residual should be down-weighted. Finally, the allocation factor maps diagnosed risk and residual trust to a small number of body-group budgets:

$$\mathcal{B}^* = \mathcal{B}_\omega(\mathcal{D}^*, \mathcal{C}^*, M_{\text{phy}}\Delta^*) \in [0, 1]^4. \quad (19)$$

The four entries correspond to root, lower-body, contact, and upper-body groups. This allocation layer bridges simulator diagnostics and the HumanML3D feature layout: frame and joint risk indicate where failure happens, while group budgets decide which representation subspaces can be safely edited. Percentile-normalized diagnostic maps are used inside $\mathcal{T}_\omega$ so that rare catastrophic failures do not dominate the scale, while local frame/joint rankings are preserved.

This factorization gives a compact answer to why multiple outputs are needed. Diagnosis alone cannot say whether the simulator residual is reliable; reliability alone cannot decide which 263D channels should be edited; and direct residual regression ignores the semantic cost of changing different body groups. By separating diagnosis, trust, and allocation, SPPM turns offline simulation into an interpretable constrained repair policy rather than a collection of ad hoc prediction heads.

Proxy architecture. SPPM predicts the same factorized decision variables from the current noised motion. Given $\mathbf{x}_t$ and timestep t , the proxy is defined as

$$F_\psi(\mathbf{x}_t, t) = (\hat{\mathcal{D}}, \hat{\mathcal{C}}, \hat{\mathcal{B}}, \hat{\Delta}). \quad (20)$$

The model uses a transformer encoder with motion tokens, positional embeddings, and timestep embeddings. Sequence-level heads predict global diagnosis, repairability, and allocation from the pooled representation, while token-level heads predict local risks, contact/root reliability, residual confidence, and residual correction. The residual head is explicitly restricted to the physical subspace:

$$\hat{\Delta} = M_{\text{phy}} \odot \tanh(H_\Delta(\mathbf{x}_t, t)). \quad (21)$$

This restriction is central: SPPM is not a second motion generator, but a physical repair proxy whose edit space is deliberately smaller than the full motion representation. Although HumanML3D features are not perfectly factorized, the chosen subspace consists of physically interpretable channels: root height, relative joint coordinates, joint velocities, and

foot contacts. The more sensitive local-rotation block is not directly edited. Thus, the correction is a small, late-stage intervention on channels that have direct physical meaning, rather than an arbitrary modification of the full 263D vector. The masked update also preserves all unselected channels exactly before VAE re-encoding:

$$\hat{\mathbf{x}} = (I - M_{\text{phy}}) \odot \mathbf{x} + M_{\text{phy}} \odot (\mathbf{x} + \mathbf{g}_\psi). \quad (22)$$

Therefore, SPPM does not freely distort the feature vector. It only proposes a clipped residual on interpretable physical coordinates, and the subsequent VAE encoding filters the edited motion back through the generator manifold.

Training objective. The proxy is trained by supervised distillation from the canonical teacher:

$$\mathcal{L}_{\text{SPPM}} = \mathcal{L}_{\text{diag}} + \lambda_\Delta \left\| W \odot (\hat{\Delta} - \Delta^*) \right\|_1 + \mathcal{L}_{\text{reg}}, \quad (23)$$

where $\mathcal{L}_{\text{diag}}$ supervises sequence quality, frame and joint risks, contact confidence, root feasibility, residual confidence, hard failure, repairability, and group budget. The residual term is not a uniform regression loss. We define its confidence map as

$$W = M_{\text{phy}} \odot \mathcal{G}_\omega(\mathcal{D}^*, \mathcal{C}^*, \mathcal{B}^*), \quad (24)$$

where $\mathcal{G}_\omega(\cdot)$ increases the weight of reliable, repairable, and group-consistent residuals and suppresses low-confidence or hard-failure regions. The regularization term penalizes non-physical leakage, unnecessary edits on semantic-sensitive components, root instability, dense residuals, contact inconsistency, and semantic drift measured by a frozen motion semantic encoder. Balanced sampling over source, motion cluster, and repair group prevents common locomotion patterns from overwhelming rare but important failure modes.

Semantic-preserved physical guidance. During inference, SPPM is activated only in the late denoising stage. At diffusion step t , the current latent $\mathbf{z}_t$ is decoded to motion space, repaired by SPPM, re-encoded, and softly blended:

$$\mathbf{z}'_t = (1 - \gamma)\mathbf{z}_t + \gamma E(\mathbf{x}_t + \mathbf{g}_\psi(\mathbf{x}_t, t, c)). \quad (25)$$

The correction is written in a factorized form:

$$\mathbf{g}_\psi = \lambda_{\text{phy}} \Gamma_\psi(\mathbf{x}_t, t, c) \odot \text{clip}(\hat{\Delta}, -\delta_{\text{max}}, \delta_{\text{max}}), \quad (26)$$

where Γ_ψ is a decision map built from predicted sequence quality, frame and joint risks, residual confidence, repairability, group budget, contact and root infeasibility, and semantic preservation gates. This factorization exposes the intended logic: SPPM first decides whether the motion is worth repairing, then where the repair should be applied, and finally how much correction can be safely injected. Semantic preservation is handled conservatively at the body-group level and then verified by a frozen motion semantic encoder. If the corrected motion deviates too far from the pre-correction motion in this semantic space, it is softly blended back toward the anchor. Thus, semantic protection is part of the reliability-aware projection rather than a separate post-hoc trick.

Why late-stage bounded repair is stable. The physical update can be understood as a local simulator projection. Suppose $\hat{\Delta} = M_{\text{phy}}(\Pi(\mathbf{x}) - \mathbf{x}) + \epsilon$ within the masked physical

Algorithm 2: Training Step-Aware Physical Proxy Model

Input: Simulator-tracker Π , official training/validation splits, motion sets $\mathcal{D}_{\text{clean}}, \mathcal{D}_{\text{hard}}, \mathcal{D}_{\text{gen}}$, SPPM F_ψ , physical mask M_{phy} , noising schedule α .

Output: Step-aware physical proxy model F_ψ .

- 1: **Offline teacher construction.** This block is run before training and cached; no simulator rollout is invoked inside the SPPM optimization loop.
- 2: Build split-safe $\mathcal{D}_{\text{phy}} = \mathcal{D}_{\text{clean}} \cup \mathcal{D}_{\text{hard}} \cup \mathcal{D}_{\text{gen}}$ from training motions/prompts.
- 3: **for** $\mathbf{x}_{\text{in}}$ in $\mathcal{D}_{\text{phy}}$ **do**
- 4: Convert $\mathbf{x}_{\text{in}}$ to an Isaac-compatible humanoid reference.
- 5: Run or load PHC-style rollout $\Pi(\mathbf{x}_{\text{in}})$ and diagnostics $\mathbf{d}$.
- 6: Convert the simulated rollout to normalized HumanML3D features $\mathbf{x}_{\text{tar}}$.
- 7: $\Delta^* \leftarrow \mathbf{x}_{\text{tar}} - \mathbf{x}_{\text{in}}$.
- 8: Build $\mathcal{Y}^* = \mathcal{T}_\omega(\mathbf{d}, \Delta^*)$ with quality, risk, confidence, reparability, budget, and residual targets.
- 9: **end for**
- 10: **SPPM training on cached canonical teacher data.**
- 11: **repeat**
- 12: **for** $(\mathbf{x}_{\text{in}}, \mathcal{Y}^*)$ in $\mathcal{D}_{\text{phy}}$ **do**
- 13: Sample timestep t and noised motion $\mathbf{x}_t \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{\text{in}}, (1 - \alpha_t)\mathbf{I})$.
- 14: Predict physical diagnosis and residual $F_\psi(\mathbf{x}_t, t)$.
- 15: Compute $\mathcal{L}_{\text{SPPM}}$ by Eq. (23) with confidence-weighted residual supervision.
- 16: $\psi \leftarrow \psi - \nabla_\psi \mathcal{L}_{\text{SPPM}}$.
- 17: **end for**
- 18: **until** converged on clean validation and deployment-style held-out monitors.

subspace, and the effective scalar step induced by Eq. (26) is $\alpha \in [0, 1]$. Let $\mathbf{e} = \mathbf{x} - \Pi(\mathbf{x})$ and $\mathbf{e}' = \mathbf{x} + \alpha M_{\text{phy}} \hat{\Delta} - \Pi(\mathbf{x})$. Under a locally fixed simulator target,

$$\|\mathbf{e}'\|_2^2 \leq (1 - \alpha)^2 \|M_{\text{phy}}\mathbf{e}\|_2^2 + \|(I - M_{\text{phy}})\mathbf{e}\|_2^2 + O(\alpha^2 \|M_{\text{phy}}\mathbf{e}\|_2^2). \quad (27)$$

Thus, when the proxy residual is accurate, the update reduces the physical residual in the repair subspace without forcing a global rewrite of the motion. Applying it in late denoising is also important: early steps mainly determine coarse semantics and global structure, while late steps are better suited for local physical correction. This yields a plug-and-play extension of ReAlign: SPM guides the sample toward semantic alignment, and SPPM performs simulator-distilled, semantic-preserved physical repair. Algorithm 3 summarizes the complete semantic-physical denoising procedure.

Experiment

Experiment Setting

Datasets and Evaluation Metrics. We employ two widely used text-to-motion datasets, HumanML3D (Guo et al.

Algorithm 3: Semantic-Physical Reward-Guided Denoising

Input: Frozen denoiser ϵ_θ , VAE encoder E and decoder D , semantic reward model R_φ , SPPM F_ψ , training set $\mathcal{D}_{\text{tr}}$, condition c , timesteps T , physical start step τ_{phy} .

Output: Generated motion $\mathbf{x}_0$.

- 1: $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: $\mathbf{x}^c = \arg \max_{\mathbf{x} \in \mathcal{D}_{\text{tr}}} R_\varphi(\mathbf{x}, c)$
- 3: **for** $t = T, \dots, 1$ **do**
- 4: Update $\mathbf{z}_{t-1}$ by the DDIM/DDPM reverse step with $\epsilon_\theta(\mathbf{z}_t, t, c)$ and CFG.
- 5: Decode $\mathbf{x}_{t-1} = D(\mathbf{z}_{t-1})$.
- 6: Compute semantic reward $R(\mathbf{x}_{t-1}, c)$ using $\mathbf{x}^c$.
- 7: Apply semantic reward guidance induced by $\nabla_{\mathbf{x}_{t-1}} R(\mathbf{x}_{t-1}, c)$ and update $\mathbf{z}_{t-1}$.
- 8: **if** $t \leq \tau_{\text{phy}}$ **then**
- 9: Decode $\mathbf{x}_{t-1} = D(\mathbf{z}_{t-1})$.
- 10: Predict SPPM diagnosis and residual $F_\psi(\mathbf{x}_{t-1}, t)$.
- 11: Build selective physical correction $\mathbf{g}_\psi$ by Eq. (26).
- 12: $\tilde{\mathbf{x}}_{t-1} \leftarrow \text{SemSafe}(\mathbf{x}_{t-1} + \mathbf{g}_\psi, \mathbf{x}_{t-1}, c)$.
- 13: $\mathbf{z}_{t-1} \leftarrow (1 - \gamma)\mathbf{z}_{t-1} + \gamma E(\tilde{\mathbf{x}}_{t-1})$.
- 14: **end if**
- 15: **end for**
- 16: **return** $D(\mathbf{z}_0)$

Method	R Precision $\uparrow$			FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
	Top 1	Top 2	Top 3			
Real	0.424	0.649	0.779	0.031	2.788	11.08
T2M (2022a)	0.361	0.559	0.681	3.022	2.052	10.72
MLD (2023b)	0.390	0.609	0.734	0.404	3.204	10.80
T2M-GPT (2023a)	0.416	0.627	0.745	0.514	3.007	10.86
CoMo (2024)	0.422	0.638	0.765	0.332	2.873	10.95
Mo.Mamba (2025b)	0.419	0.645	0.765	0.307	3.021	11.02
ParCo (2025)	0.430	0.649	0.772	0.453	2.820	10.95
Mo.Diffuse (2024)	0.417	0.621	0.739	1.954	2.958	11.10
w/ ReAlign (Ours)	0.419	0.639	0.764	0.805	2.801	10.66
MDM (2023)	0.403	0.606	0.731	0.497	3.096	10.74
w/ ReAlign (Ours)	0.451	0.664	0.784	0.276	2.775	10.76

Table 1: **Comparison of text-to-motion generation performance on the KIT-ML dataset.** Bold highlights the best results except for Diversity, which is reported for reference.

2022a) and KIT-ML (Plappert, Mandery, and Asfour 2016), following their official train/validation/test splits. HumanML3D motions are represented as 20 FPS 263D feature sequences with lengths in the range of 40–200 frames in our MLD-based experiments. Consistent with prior work (Guo et al. 2024; Li et al. 2025c), we adopt R-Precision for Top k , Fréchet Inception Distance (FID), Matching score, and Diversity to assess semantic generation quality. For the HumanML3D physical analysis, we report raw physical attributes following the CLoSD text-to-motion protocol and further evaluate post-simulation executability by re-playing each motion with the same PHC/Isaac humanoid tracker. We use penetration, floating, skating, GC-W1, and Contact F1 to analyze geometric and contact quality, and use reset early, PHC success rate, tracking error, Jerk20, and post-simulation floating/skating to analyze physical executability. Because raw floor-contact attributes can be sensi-

tive to method-specific height bias and coordinate recovery, we treat them as geometric diagnostics and use PHC/Isaac tracking metrics as the primary evidence for physical executability.

Implementation Details. We employ the SkipTransformer (Chen et al. 2023b) as the foundational architecture for our step-aware semantic reward model, consisting of a transformer encoder processing both text and motion inputs, alongside a motion decoder. Each component features 9 layers and 4 attention heads, with the latent space dimension fixed at 256. The training process incorporates a maximum timestep of 1000, a noisy motion probability of 0.5, and a negative filtering threshold of 0.9 to regulate the selection of negative samples. For model training, we adhere to the TMR framework (Petrovich, Black, and Varol 2023), employing a composite loss function expressed as a weighted combination $\mathcal{L}_C + \mathcal{L}_R$. Optimization is performed using the AdamW algorithm (Loshchilov and Hutter 2017), configured with a learning rate of 10^{-4} and a batch size of 512, while other hyperparameters are consistent with those specified in the TMR (Petrovich, Black, and Varol 2023). Unless otherwise specified, MLD-based generation uses the frozen public MLD checkpoint, 50-step DDIM sampling, classifier-free guidance with scale 7.5 and unconditional probability 0.1. The semantic reward branch uses the trained SPM checkpoint with $\lambda_{t2m} = \lambda_{m2m} = 100$ for the semantic-only ReAlign setting. We denote the semantic-preserved physical-proxy variant as ReAlign-Phy.

SPPM and Physical Evaluation Setup. The physical proxy is trained only from cached offline teacher data and is never coupled to an online simulator during diffusion sampling. Its training set contains 16,170 canonical teacher samples: 9,749 clean reference replays, 1,838 hard replay samples, and 4,583 generated-motion replay samples, all constructed from training motions or training prompts. Checkpoint selection uses 613 clean validation samples together with two deployment-style held-out monitors containing 204 hard-replay samples and 510 generated-replay samples. SPPM uses a 4-layer transformer with hidden dimension 256, 4 attention heads, maximum length 256, dropout 0.1, and sinusoidal timestep embeddings. It is trained with batch size 256, AdamW, learning rate 1.5×10^{-4} , cosine decay, 5 warmup epochs, weight decay 10^{-4} , and timestep-dependent Gaussian input perturbation with base standard deviation 0.03. Training is run for at most 100 epochs and early stops after 71 epochs; the best checkpoint is selected at epoch 63 by a composite validation score over the clean and held-out physical monitors.

During generation, SPPM is activated only in the last 40% of the 50 denoising steps. We set $\lambda_{sppm} = 0.25$, latent blending coefficient $\gamma = 0.18$, residual clipping threshold $\delta_{max} = 0.25$, and risk exponent 1.5 for ReAlign-Phy. Physical evaluation is performed after generation on saved motions, so all methods are compared under the same conversion and simulation pipeline. Each motion is converted from 20 FPS HumanML3D output to a 30 FPS PHC/Isaac-compatible humanoid reference, fitted with 100 SMPLify iterations, and replayed by the same PHC-style controller in headless mode. We use the axis-angle tracking pipeline

with no extra joint-order conversion, inverse root-orientation prior, zero additional yaw offsets, 12 workers per job, and four GPUs for parallel evaluation. For the semantic-physical tables, ReAlign-Phy denotes our physical proxy-enhanced ReAlign variant. Diversity is reported as a reference distributional metric.

Results of Motion Generation and Retrieval

Standard Text-to-Motion Generation. Table 2 preserves the original ReAlign comparison on the standard HumanML3D text-to-motion benchmark. The semantic reward-guided sampler remains a plug-and-play improvement over diffusion backbones: when applied to MLD, ReAlign improves R@1/R@2/R@3 from 0.481/0.673/0.772 to 0.567/0.759/0.848, reduces FID from 0.473 to 0.195, and lowers MM Dist from 3.196 to 2.704. The same trend holds for MLD++, where ReAlign further improves all alignment metrics and reaches the strongest reported scores in this table. This standard benchmark establishes the semantic alignment contribution of the original step-aware reward guidance; the following tables analyze whether the physically extended variant preserves this benefit while improving executability.

Unified Semantic-Physical Evaluation. Table 3 connects semantic generation quality with downstream physical replay. MLD w/ ReAlign-Phy keeps strong text-motion alignment, achieving 0.570/0.760/0.850 in R@1/R@2/R@3 and the best Matching score of 2.710 among generated methods. More importantly, it substantially improves physical executability: compared with MLD, it reduces Jerk20 from 114.80 to 60.24, reduces Track MPJPE from 81.31 to 54.96, and improves Track SR from 80.9% to 89.3%. Compared with MLD w/ ReAlign, ReAlign-Phy turns the semantic alignment gain into a more executable motion by reducing Track MPJPE from 79.62 to 54.96 and Jerk20 from 112.52 to 60.24. This is the intended role of SPPM: it does not replace semantic guidance, but adds a bounded physical repair step after the text-conditioned structure has formed.

Comparison with Physics-related Generation Methods. Table 4 places MLD w/ ReAlign-Phy next to representative physics-related motion generation rows. We keep simulator-based systems directly below the Real reference and report non-simulator generators separately. MoConVQ is kept as a reported CLoSD reference, while CLoSD provides an executed-motion reference produced with an online controller. Under the non-simulator group, ReAlign-Phy obtains the best Matching, R@1, and R@3 among generated methods, while reducing floating from 24.832 mm for MLD to 22.953 mm and skating from 1.068 mm to 0.477 mm. DiP has the lowest penetration and skate ratio, but ReAlign-Phy provides the strongest balance between semantic alignment and raw floating/skating quality.

Supplementary Physical Diagnostics. Table 5 complements the raw attributes with clearance, self-contact, and contact-distribution diagnostics. ReAlign-Phy obtains the best generated-method Float P95, SelfContact coverage, SelfContact Skate, HCS-AUC, GC-W1, and Contact F1. Compared with MLD, it reduces Float P95 from 82.930 mm to 67.409 mm and GC-W1 from 4.416 to 2.540, while in-

Method	R Precision $\uparrow$			FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
	Top 1	Top 2	Top 3			
Real	0.511	0.703	0.797	0.002	2.974	9.503
T2M (2022a)	0.455 $\pm$ 0.002	0.636 $\pm$ 0.003	0.736 $\pm$ 0.003	1.087 $\pm$ 0.002	3.347 $\pm$ 0.008	9.175 $\pm$ 0.002
MDM (2023)	0.455 $\pm$ 0.006	0.645 $\pm$ 0.007	0.749 $\pm$ 0.006	0.489 $\pm$ 0.047	3.330 $\pm$ 0.25	9.920 $\pm$ 0.083
T2M-GPT (2023a)	0.492 $\pm$ 0.003	0.679 $\pm$ 0.002	0.775 $\pm$ 0.002	0.141 $\pm$ 0.005	3.121 $\pm$ 0.009	9.722 $\pm$ 0.082
ReMoDiffuse (2023b)	0.510 $\pm$ 0.005	0.698 $\pm$ 0.006	0.795 $\pm$ 0.004	0.103 $\pm$ 0.004	2.974 $\pm$ 0.016	9.018 $\pm$ 0.75
Mo.Diffuse (2024)	0.491 $\pm$ 0.001	0.681 $\pm$ 0.001	0.775 $\pm$ 0.001	0.630 $\pm$ 0.001	3.113 $\pm$ 0.001	9.410 $\pm$ 0.049
OMG (2024)	-	-	0.784 $\pm$ 0.002	0.381 $\pm$ 0.008	-	9.657 $\pm$ 0.085
MotionLCM (2024)	0.502 $\pm$ 0.003	0.698 $\pm$ 0.002	0.798 $\pm$ 0.002	0.304 $\pm$ 0.012	3.012 $\pm$ 0.007	9.607 $\pm$ 0.066
Mo.Mamba (2025b)	0.502 $\pm$ 0.003	0.693 $\pm$ 0.002	0.792 $\pm$ 0.002	0.281 $\pm$ 0.011	3.060 $\pm$ 0.000	9.871 $\pm$ 0.084
CoMo (2024)	0.502 $\pm$ 0.002	0.692 $\pm$ 0.007	0.790 $\pm$ 0.002	0.262 $\pm$ 0.004	3.032 $\pm$ 0.015	9.936 $\pm$ 0.066
ParCo (2025)	0.515 $\pm$ 0.003	0.706 $\pm$ 0.003	0.801 $\pm$ 0.002	0.109 $\pm$ 0.005	2.927 $\pm$ 0.008	9.576 $\pm$ 0.088
MARDM (2024)	0.500 $\pm$ 0.004	0.695 $\pm$ 0.003	0.795 $\pm$ 0.003	0.114 $\pm$ 0.007	-	-
MG-MotionLLM (2025b)	0.516 $\pm$ 0.002	0.706 $\pm$ 0.002	0.802 $\pm$ 0.003	0.303 $\pm$ 0.010	2.952 $\pm$ 0.009	9.960 $\pm$ 0.073
EnergyMoGen (2025)	0.526 $\pm$ 0.003	0.718 $\pm$ 0.003	0.815 $\pm$ 0.002	0.176 $\pm$ 0.006	2.931 $\pm$ 0.007	9.500 $\pm$ 0.091
MLD (2023b)	0.481 $\pm$ 0.003	0.673 $\pm$ 0.003	0.772 $\pm$ 0.002	0.473 $\pm$ 0.013	3.196 $\pm$ 0.010	9.724 $\pm$ 0.082
w/ ReAlign (Ours)	0.567 $\pm$ 0.003 (+17.9%)	0.759 $\pm$ 0.003 (+12.8%)	0.848 $\pm$ 0.003 (+9.8%)	0.195 $\pm$ 0.005 (+58.8%)	2.704 $\pm$ 0.007 (+15.4%)	9.474 $\pm$ 0.068 (+86.9%)
MLD++(2024)	0.548 $\pm$ 0.003	0.738 $\pm$ 0.003	0.829 $\pm$ 0.002	0.073 $\pm$ 0.003	2.810 $\pm$ 0.008	9.658 $\pm$ 0.089
w/ ReAlign (Ours)	0.572 $\pm$ 0.002 (+4.4%)	0.764 $\pm$ 0.002 (+3.5%)	0.852 $\pm$ 0.001 (+2.8%)	0.055 $\pm$ 0.003 (+24.7%)	2.648 $\pm$ 0.008 (+5.8%)	9.478 $\pm$ 0.055 (+83.9%)

Table 2: **Comparison of text-to-motion generation performance on the HumanML3D dataset.** These metrics are evaluated by the evaluator from TM2T (Guo et al. 2022b). The arrows $\uparrow$, $\downarrow$, and $\rightarrow$ indicate higher, lower, and closer-to-real-motion values are better, respectively. **Bold** highlights the best results except for Diversity, which is reported for reference. Percentages in bracket indicate improvements over respective baselines.

Method	Semantic Generation Metrics						Key Post-simulation Metrics		
	FID $\downarrow$	Matching $\downarrow$	R@1 $\uparrow$	R@2 $\uparrow$	R@3 $\uparrow$	Diversity $\rightarrow$	Jerk20 $\downarrow$	Track MPJPE $\downarrow$	Track SR $\uparrow$
Real	0.000	2.986	0.516	0.699	0.793	9.520	58.42	50.09	90.7
MLD	0.392	3.022	0.494	0.685	0.790	9.650	114.80	81.31	80.9
MDM	0.405	3.614	0.417	0.603	0.705	9.456	77.28	62.90	86.0
DiP	0.190	2.930	0.521	0.713	0.812	9.773	<u>61.89</u>	<u>55.83</u>	<u>88.0</u>
MLD w/ ReAlign	<u>0.315</u>	<u>2.811</u>	<u>0.546</u>	<u>0.738</u>	<u>0.833</u>	9.411	112.52	79.62	81.0
MLD w/ ReAlign-Phy (Ours)	0.388	2.710	0.570	0.760	0.850	9.910	60.24	54.96	89.3

Table 3: **Unified HumanML3D semantic and post-simulation analysis.** “MLD w/ ReAlign” denotes the semantic reward-guided sampler, and “MLD w/ ReAlign-Phy (Ours)” denotes our physical proxy-enhanced ReAlign variant. Bold marks the best generated method; underline marks the second best. Diversity is reported for reference.

creasing Contact F1 from 0.7076 to 0.7477. These results show that the physical proxy improves not only average floor-contact geometry but also tail floating, self-contact stability, and contact distribution.

Motion-Text Retrieval. Following the small-batch protocol of TMR(Petrovich, Black, and Varol 2023), we evaluate our ReAlign on text–motion retrieval and benchmark it against the latest state of the art. As summarized in Table 6, ours attains 67.59% R@1 and 87.44% R@3 on HumanML3D for the text-to-motion retrieval, and reaches 68.94% R@1 and 82.86% R@2 in the motion-to-text retrieval, consistently surpassing LaMP (Li et al. 2025c) and TMR(Petrovich, Black, and Varol 2023). On KIT-ML, our approach further pushes performance to 91.19% R@5 and 84.38% R@3, consistently surpassing baselines. We attribute these improvements to our noise augmentation strategy, which alleviates the limited motion diversity and text annotations in both datasets that lead to many hard negative samples, thereby enhancing the model’s discriminative capability for subtle motion variations.

Plug-and-Play Capability of ReAlign. To demonstrate the plug-and-play capability and generalizability of our Re-

Align, we integrate it into various diffusion-based models for text-to-motion generation, as shown in Table 7. Across methods such as Mo.Diffuse (Zhang et al. 2024), MDM (Tevet et al. 2023), MLD (Chen et al. 2023b), MotionLCM and its extension MLD++ (Dai et al. 2024). Our ReAlign consistently enhances performance. Notably, it achieves substantial improvements in alignment quality and motion realism, with relative gains of up to 17.9% in R@1 and 58.8% in FID for MLD. While diversity slightly decreases in some cases, this is expected and beneficial. Better diversity does not always indicate better quality, as it simply reflects motion variety. ReAlign prioritizes well-aligned motions over misaligned ones, leading to significant gains in other metrics without compromising generation quality. These results underscore the plug-and-play capability of this module, effectively elevating the efficacy of diverse motion generation frameworks.

Ablation Studies and Discussions

Effectiveness of Handling Noisy. To verify the necessity of handling noise and to avoid reward hacking, we varied the denoising steps of MLD from 1 to 50, employing

Generation Process	Method	Semantic Metrics						Physical Attributes			
		FID↓	Matching↓	R@1↑	R@2↑	R@3↑	Diversity→	Penetration↓[mm]	Floating↓[mm]	SkateRatio↓	Skating↓[mm]
Reference	Real HumanML3D GT	0.000	2.986	0.516	0.699	0.793	9.520	0.000	22.812	0.059	0.207
w/ physics simulator	MoConVQ (Yao et al. 2024)	3.279	3.970	0.309	0.504	0.614	8.010	0.249	32.000	-	0.294
	CLoSD (Tevet et al. 2025)	1.541	3.490	0.419	0.606	0.724	8.439	0.027	19.652	0.026	0.001
w/o physics simulator	MLD (Chen et al. 2023b)	0.392	3.022	0.494	0.685	0.790	9.650	1.026	24.832	0.086	1.068
	MDM (Tevet et al. 2023)	0.405	3.614	0.417	0.603	0.705	9.456	1.031	26.834	0.085	1.425
	DiP	0.190	2.930	0.521	0.713	0.812	9.773	0.117	24.640	0.057	0.660
	MLD w/ ReAlign	0.315	2.811	0.546	0.738	0.833	9.411	1.091	24.771	0.088	1.112
	MLD w/ ReAlign-Phy (Ours)	0.388	2.710	0.570	0.760	0.850	9.910	<u>0.405</u>	22.953	<u>0.061</u>	0.477

Table 4: **Comparison with representative physics-related motion generation results.** “MLD w/ ReAlign-Phy (Ours)” denotes our physical proxy-enhanced ReAlign variant. MoConVQ is not reproduced and follows the value reported by CLoSD; CLoSD is kept as a simulator-based executed-motion reference. Rows with a physics simulator are placed directly below Real because they produce executed motions during generation; non-simulator rows are reported separately below them. Bold marks the best method inside each generation-process group; underline marks the second best. Diversity is reported for reference.

Method	Float P95↓	SelfContact Cov↑	SelfContact Skate↓	HCS-AUC↓	GC-W1↓	Contact F1↑
Real HumanML3D GT	72.968	0.7479	2.270	1.590	0.000	1.0000
MLD	82.930	0.6913	5.338	2.900	4.416	0.7076
MDM	108.521	0.6938	4.224	3.477	11.348	0.6914
DiP	<u>75.193</u>	<u>0.7398</u>	4.741	<u>2.720</u>	45.796	0.4847
MLD w/ ReAlign	79.978	0.6948	5.369	2.948	<u>4.219</u>	<u>0.7102</u>
MLD w/ ReAlign-Phy (Ours)	67.409	0.7504	4.054	2.069	2.540	0.7477

Table 5: **Supplementary physical diagnostics on HumanML3D.** Penetration, floating, and skating are reported in Table 4; this table keeps the complementary clearance, self-contact, and contact-distribution diagnostics. “MLD w/ ReAlign-Phy (Ours)” denotes our physical proxy-enhanced ReAlign variant. Bold marks the best generated method; underline marks the second best.

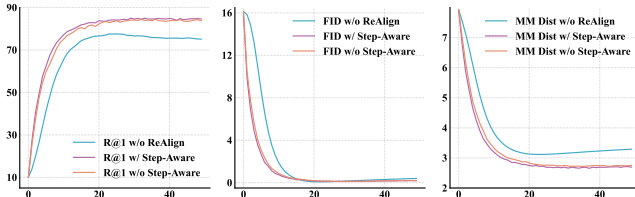


Figure 4: Comparison of motion generation quality across denoising steps for the MLD w/o ReAlign, MLD w/o Step-Aware, and MLD w/ Step-Aware (ReAlign). ReAlign consistently outperforms the others, highlighting the necessity of explicit noise handling during denoising.

both the reward model (RM) and the ReAlign to perform reward-guided sampling. This compels the model to generate noisy motions. As shown in Figure 4, compared to the baseline and the RM, ours achieves superior performance across all denoising steps, demonstrating that explicitly handling noise during the denoising benefits the generation of higher quality motions. Notably, we observed that MLD generates motions with discernible semantics even in the early steps of denoising, rather than purely noise. This behavior may be attributed to the latent VAE used in MLD, which inherently exhibits robustness to noise. These experimental results demonstrate the feasibility and necessity of handling noise in denoising, aligning with conclusions drawn in DALLE-2 (2022) and GLIDE (2021).

Effectiveness of Reward Sampling. We assess the ReAlign, including T2M and M2M alignment rewards, along with the step-aware strategy in text-to-motion generation on MLD (Chen et al. 2023b). As shown in Table 8, results indicate that the T2M reward significantly improves the alignment between the generated motions and text descriptions, as well as the realism of the motions. While the M2M reward alone exhibits limited efficacy due to the inaccuracy of text-to-motion retrieval, its integration with the step-aware strategy further enhances motion realism, validating the necessity of handling noise during the sampling. The combination of T2M and step-aware strategies achieves optimal performance, with M2M providing additional realism gains.

Effectiveness of Physical Proxy Repair. Table 9 separates the semantic reward branch from the physical proxy branch. ReAlign alone improves semantic alignment over MLD, reducing Matching from 3.022 to 2.811 and increasing R@1 from 0.494 to 0.546, but it only mildly improves physical replay. SPPM alone produces the opposite behavior: it improves raw physical metrics and post-simulation tracking, reducing GC-W1 from 4.416 to 2.278 and Track MPJPE from 81.31 to 57.08, while the semantic scores are not optimal. Combining the two branches gives the desired trade-off. ReAlign-Phy obtains the best Matching, floating, PHC success rate, and Track MPJPE in the ablation, confirming that semantic reward guidance and simulator-distilled physical repair are complementary.

Discussion on ReAlign and Classifier-free Guidance. As shown in Table 10, our ReAlign is compatible with

	Methods	Noise	Text-Motion Retrieval $\uparrow$					Motion-Text Retrieval $\uparrow$				
			R@1	R@2	R@3	R@5	R@10	R@1	R@2	R@3	R@5	R@10
HumanML3D	TEMOS (2022)	$\times$	40.49	53.52	61.14	70.96	84.15	39.96	53.49	61.79	72.40	85.89
	T2M (2022b)	$\times$	52.48	71.05	80.65	89.66	96.58	52.00	71.21	81.11	89.87	<u>96.78</u>
	TMR (2023)	$\times$	67.16	81.32	86.81	91.43	95.36	67.97	81.20	86.35	91.70	95.27
	LaMP (2025c)	$\times$	67.18	81.90	87.04	92.00	95.73	68.02	82.10	87.50	92.20	96.90
	ReAlign (ours)	$\checkmark$	67.59	82.24	87.44	<u>91.97</u>	<u>96.28</u>	68.94	82.86	87.95	92.44	<u>96.28</u>
KIT-ML	TEMOS (2022)	$\times$	43.88	58.25	67.00	74.00	84.75	41.88	55.88	65.62	75.25	85.75
	T2M (2022b)	$\times$	42.25	62.62	75.12	87.50	96.12	39.75	62.75	73.62	86.88	95.88
	TMR (2023)	$\times$	49.25	69.75	78.25	87.88	95.00	50.12	67.12	76.88	88.88	94.75
	ReAlign (ours)	$\checkmark$	52.84	71.66	82.96	91.19	97.59	52.98	72.87	84.38	92.61	96.87

Table 6: **Comparison of Text-to-motion (left) and motion-to-text (right) retrieval methods on the HumanML3D and KIT-ML datasets.** The column “Noise” indicates whether the method can handle noisy motion from the denoised process.

Method	R Precision $\uparrow$			FID $\downarrow$	MM Dist $\downarrow$ Diversity $\rightarrow$	
	Top 1	Top 2	Top 3			
Real	0.511	0.703	0.797	0.002	2.974	9.503
MDiff (2024)	0.491	0.681	0.775	0.630	3.113	9.410
w/ ReAlign	0.534 _{+8.8%}	0.733 _{+7.6%}	0.829 _{+7.0%}	0.370 _{+41.3%}	2.807 _{+9.9%}	9.372 _{+0.04%} <u>96.9%</u>
MDM (2023)	0.455	0.645	0.749	0.489	3.330	9.920
w/ ReAlign	0.470 _{+3.3%}	0.677 _{+5.0%}	0.789 _{+5.3%}	0.325 _{+33.5%}	3.129 _{+6.0%}	9.355 _{+0.27%} <u>64.5%</u>
MLD (2023b)	0.481	0.673	0.772	0.473	3.196	9.724
w/ ReAlign	0.567 _{+17.9%}	0.759 _{+12.8%}	0.848 _{+9.8%}	0.195 _{+58.8%}	2.704 _{+15.4%}	9.474 _{+0.19%} <u>86.9%</u>
MLCM ⁴ (2024)	0.502	0.698	0.798	0.304	3.012	9.607
w/ ReAlign	0.540 _{+7.6%}	0.739 _{+5.9%}	0.833 _{+4.4%}	0.273 _{+10.2%}	2.797 _{+7.1%}	9.683 _{+0.08%} <u>73.1%</u>
MLD++ (2024)	0.548	0.738	0.829	0.073	2.810	9.658
w/ ReAlign	0.572 _{+4.4%}	0.764 _{+3.5%}	0.852 _{+2.8%}	0.055 _{+24.7%}	2.648 _{+5.8%}	9.478 _{+0.13%} <u>83.9%</u>

Table 7: **Performance improvement of motion generation methods with our step-aware reward guidance.** Results are reported on the HumanML3D dataset, showing improvements over baseline methods.

Classifier-Free Guidance (CFG) (Ho and Salimans 2022), and their integration can further unleash the performance of the diffusion model. Unlike CFG, which requires training, our ReAlign is plug-and-play and supports flexible reward function design tailored to different tasks (e.g., physical reward, trajectory reward, style reward, etc.). In this work, we primarily focus on improving text-motion alignment, while future research will explore reward designs for more tasks.

Conclusion

We propose ReAlign, a plug-and-play reward-guided sampling strategy for diffusion-based text-to-motion generation. By jointly optimizing text-aligned and motion-aligned rewards during denoising, ReAlign effectively improves semantic consistency and motion realism. Our method integrates seamlessly with existing diffusion models without extra fine-tuning. Extensive experiments demonstrate that ReAlign achieves significant gains in both text-motion alignment and motion quality over state-of-the-art baselines.

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T2M	M2M	SA	R Precision $\uparrow$			FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
			Top 1	Top 2	Top 3			
$\times$	$\times$	$\times$	0.481	0.673	0.772	0.473	3.196	9.724
$\checkmark$	$\times$	$\times$	0.556	0.747	0.841	0.213	2.761	9.516
$\times$	$\checkmark$	$\times$	0.517	0.721	0.809	0.205	2.932	9.455
$\checkmark$	$\checkmark$	$\times$	0.556	0.750	0.840	<u>0.199</u>	2.750	9.529
$\checkmark$	$\times$	$\checkmark$	0.568	0.761	0.850	0.212	<u>2.714</u>	9.598
$\times$	$\checkmark$	$\checkmark$	0.523	0.709	0.810	0.203	2.963	9.525
$\checkmark$	$\checkmark$	$\checkmark$	<u>0.567</u>	<u>0.759</u>	<u>0.848</u>	0.195	2.704	9.474

Table 8: **Ablation study of the text-to-motion on HumanML3D dataset.** “T2M”, “M2M” and “SA” denote the text-to-motion reward, motion-to-motion reward and step-aware training, respectively. Diversity is reported for reference.

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Method	SPM	SPPM	Semantic		Raw Physical			Post-simulation		
			Matching↓	R@1↑	Floating↓	GC-W1↓	Contact F1↑	PHC SR↑	Track MPJPE↓	
MLD	✗	✗	3.022	0.494	24.832	4.416	0.708	80.9	81.31	
MLD w/ ReAlign	✓	✗	<u>2.811</u>	<u>0.546</u>	24.771	4.219	0.710	81.0	79.62	
MLD w/ SPPM	✗	✓	2.909	0.530	<u>23.332</u>	2.278	<u>0.744</u>	<u>88.0</u>	<u>57.08</u>	
MLD w/ ReAlign-Phy (Ours)	✓	✓	2.710	0.570	22.953	<u>2.540</u>	0.748	89.3	54.96	

Table 9: **Ablation of semantic reward guidance and the physical proxy on HumanML3D.** SPM denotes the step-aware semantic reward model, and SPPM denotes the step-aware physical proxy model. “MLD w/ ReAlign-Phy (Ours)” denotes the full semantic-physical variant. Bold marks the best generated method; underline marks the second best.

CFG	ReAlign	R Precision ↑			FID ↓	MM Dist ↓	Diversity →
		Top 1	Top 2	Top 3			
✗	✗	0.263	0.407	0.506	0.586	4.823	8.613
✓	✗	0.481	0.673	0.772	0.473	3.196	9.724
✓	✓	0.567	0.759	0.848	0.195	2.704	9.474

Table 10: **Ablation study of the guidance strategy.** Evaluation conducted on the HumanML3D with MLD (2023b) as the baseline. “CFG” and “ReAlign” denote the classifier-free guidance and our ReAlign, respectively. Diversity is reported for reference.

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